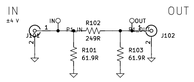
P1 pi pad, −20 dB

tutorial 6.12, 5.06

cell (0,0)

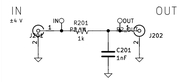


EXPECT: −20 dB flat, 50 ohm both ways
A pi attenuator: 61.9 / 249 / 61.9 ohm is −20.0 dB and 50.0 ohm in a 50 ohm system (6.12's $10 pad, on the board).
Into a high-impedance load it is only −14 dB: the pad assumes 50 ohm on both sides.

P2 RC low-pass, 159 kHz

tutorial 4.02, 4.08, 7.06

cell (1,0)

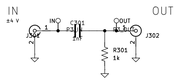


EXPECT: −3 dB at 159 kHz, then −20 dB/decade; OUT into the ADC's high impedance
1 k and 1 nF: f = 1 / (2 pi R C) = 159 kHz. Measure OUT with the ADC on its high-impedance setting: a 50 ohm
load would make a 1 k / 50 divider of it. The lag plant of 4.08 and 7.06: the step response is 1 − exp(−t / 1 us).

P3 RC high-pass, 159 kHz

tutorial 4.02

cell (2,0)

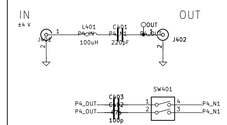


EXPECT: −3 dB at 159 kHz, +20 dB/decade below; OUT into high impedance
The mirror image of P2: the same 1 k and 1 nF the other way round. A step in gives a spike that decays in 1 us.

P4 series RLC, 100uH 220pF

tutorial 4.02, 7.06

cell (3,0)



the DIP switch and its two capacitors are drawn on the right: way 1 adds 47 pF, way 2 adds 100 pF across C
(each way from the L–C node to the OUT node through its capacitor)

EXPECT: a peak at 1.07 MHz (switches off), 0.98 with 1, 0.89 with 2, 0.83 with both; Q about 6
Series L C between 50 ohm ports: a band-pass at f = 1 / (2 pi sqrt(L C)) whose Q ≈ sqrt(L/C) / (100 + R_coil) is about 6
(the 10 ohm coil would give 60 by itself). The DIP switch adds 47 pF (way 1) and 100 pF (way 2) across the 220 pF to move
the peak: a resonant plant for 7.06, a lockbox's cavity before the real one. No IN loop (the switch took its corner).

P5 LC low-pass, 9 MHz

tutorial 1.06, 6.10, 4.02

cell (0,1)

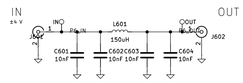


EXPECT: −3 dB at 8.9 MHz, −30 dB at 18 MHz: the anti-alias filter in front of a 25 MS/s ADC
5th-order Butterworth, 50 ohm: 220 pF, 1.5 uH, 680 pF, 1.5 uH, 220 pF (the exact values are 219 pF, 1.43 uH, 707 pF).
Before the ADC it removes 6.10's ghost; after the DAC it smooths the steps.

P6 LC low-pass, 100 kHz

tutorial 7.05, 4.02

cell (1,1)

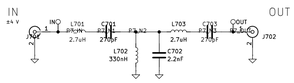


EXPECT: −3 dB at 118 kHz: the reconstruction filter after a one-bit DAC
3rd-order pi, 50 ohms: 2 x 10 nF, 150 uH, 2 x 10 nF, all C0G (JLCPCB stocks no 22 nF C0G in 0805 or 1206).
Turns 7.05's one-bit stream into a tone: the slow plant of 7.06 with P2.

P7 band-pass, 4.7–7.5 MHz

tutorial 6.10, 5.07, 6.12

cell (2,1)

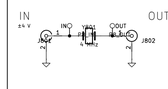


EXPECT: Chapter 6's band: a peak at 6.3 MHz, 3 dB from 4.7 to 7.5 MHz
3rd-order Butterworth band-pass, 6.05 MHz centre, 3.1 MHz wide: series 2.7 uH + 270 pF, shunt 330 nH || 2.2 nF, series 2.7 uH + 270 pF
(the exact values: 2.57 uH, 269 pF, 337 nH, 2.05 nF). 6.10's filtered chips in copper; the transmitter's harmonics and the
receiver's front end of 5.07 and 6.12.

P8 quartz crystal, 4 MHz

tutorial 4.03, 4.09, 4.10

cell (3,1)



EXPECT: a sharp pass at the series resonance near 4.000 MHz (−3.7 dB) and a notch 6.0 kHz above it, the parallel resonance
The crystal in series between 50 ohm ports: it passes at its series resonance f_s (R_m about 50 ohm: −3.5 dB) and blocks at
the parallel resonance f_p = f_s (1 + C_m / 2 C_0) a few kHz above. A capacitor across OUT (a tee at the ADC) pulls f_p
down toward f_s and leaves f_s alone: the pulling that 4.09 measures. Model: C_m 15 fF, Q 50 000, C_0 4 pF (typical HC-49S).

P9 ceramic res., 455 kHz

tutorial 4.02, 4.03

cell (0,2)

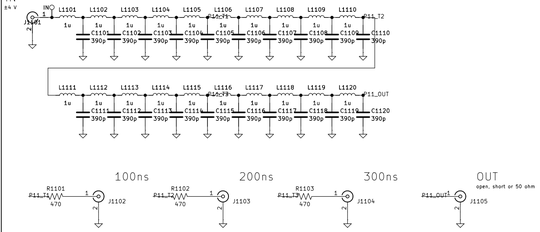


EXPECT: a pass near 455 kHz and a notch a few kHz above: a resonator with a Q of a few hundred, between P4's LC and P8's quartz
A two-terminal 455 kHz ceramic resonator in series (JLCPCB stocks no three-terminal 455 kHz filter). It is the superhet's IF
element, and here a resonance whose Q (hundreds) sits between the LC's and the crystal's. Model: C_m 7 pF, Q 400, C_0 250 pF.

P11 delay line, 400 ns, a tap every 100

tutorial 4.05, 6.05, 6.10, 7.06

cells (1,2), (2,2), (3,2)

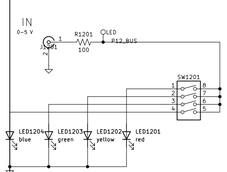


EXPECT: a pulse arrives at OUT 395 ns late, at the taps every 99 ns; flat to about 10 MHz, then it disperses
Twenty sections of 1.0 uH and 390 pF: Z0 = sqrt(L/C) = 51 ohm, 20 ns a section, cut-off 1 / (pi sqrt(LC)) = 16 MHz,
400 ns is 260 ft of RG-58. Leave OUT open and a pulse comes back the same way up; short it and it comes back inverted:
50 ohm on OUT and nothing comes back (4.05). The taps are 6.05's echo and 6.10's radar target: near 16 MHz the
phase velocity falls and a chirp spreads (7.06's pure-delay plant below that). A tap through 470 ohms: x1 into a
high-impedance scope, /10 into 50 ohm.

P12 LEDs

tutorial 4.10

cell (0,3)

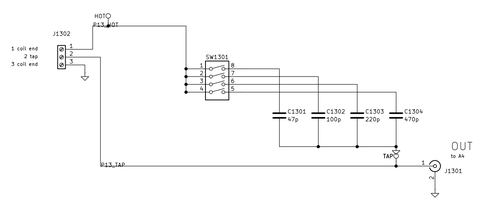


EXPECT: one LED at a time: the voltage where it starts to glow against I_sun·wavelength is h/c
100 ohm from the jack to the four LEDs; the 4-way switch picks one. Drive from the DAC through the ECP5 board's unipolar
LED output or a bench supply: the knee voltage of each colour against c / lambda gives Planck's constant (4.10).
Way 1 red 630 nm (AlGaInP), 2 yellow 596 (AlGaInP), 3 green 525 (InGaN), 4 blue 469 nm (InGaN): the InGaN pair lie
above the line. The LED loop is after the 100 ohm: the LED's own voltage. The colours are not on the silk: light them.

P13 AM loopstick, 0.5–1.7 MHz

tutorial 1.10

cells (1,3), (2,3)

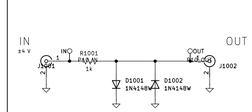


EXPECT: with a 330 uH loopstick: 1.28 MHz on 1, 0.88 on 2, 0.59 on 3, 0.40 MHz on 4, and the sums between (the AM band)
Any ferrite-rod AM antenna coil (about 330 uH: 50 to 100 mm long) screws into the terminal. Its tap (a quarter of the turns
from the ground end) to the jack. The switch's capacitors tune it: f = 1 / (2 pi sqrt(L C)). 1.28 MHz with 47 pF down to
0.40 MHz with 470 pF; combine positions for the steps between. Into AA's LNA and 1.10's AM radio: a real station.

P10 clipper, 1N4148 ±0.6 V

tutorial 4.04

cell (3,3)



EXPECT: a sine of more than 1 V comes out flat-topped at about ±0.6 V; the ADC on high impedance
1 k in series, then a 1N4148W each way to ground: whatever goes in, OUT stays within a diode drop. 4.04's harmonics: the
clipped sine's spectrum against the amplitude; the 1 k limits the diode current to a few milliamps from a ±4 V source.

MH1 MH2 MH3 MH4
M3 mounting holes, one per corner

GND GND GND GND
ground loops for the scope's clips, one per channel, in the board's margins

PWR_FLAG

H9901 H9902 H9903 H9904
slots for cable ties

Drawn by scripts/gen_sch.py from boards/amod_p.py: the expected responses from sim/passive.py
Every tile is one circuit with its own SMA jacks; the board's silkscreen draws each tile's wires as they
pads. RC and LC filters, resonators, a crystal, a clipper, a tapped delay line, LEDs, an AM loopstick
Board: Jason Gallicchio and Claude | open hardware, CERN-OHL-P-2.0 | github.com/jgallicchio/amod

Sheet: /
File: amod_p.kicad_sch

Title: AMOD-P: passive building blocks for the converters

Size: A1	Date: 2026-10-25	**Rev: A**
KiCad E.D.A. 10.0.7		Id: 1/1